Delta Audit Result — Paper III v1.1.6

Narrow editorial-delta re-audit against the validated v1.1.5 package

DELTA VERDICT · PASS

Paper	Linear-Error Clique Partitions of Split Graphs via Structured Triangle Packing (Paper III, Erd■s #81)
Baseline	v1.1.5 — audited PASS_WITH_OBSERVATIONS (sole open item F-G05, minor/editorial)
This delta	v1.1.6 — editorial-only; resolves F-G05
Manuscript v1.1.5	7aaf03083ddf7731dcb2b1e849cdfac97fb1697df1650c49a56e8431ce1bcb0b
Manuscript v1.1.6	b54d0b79c2c2a4ff21a1fe16f388de771a261c51fab0f6cfceea53559102e55d
Auditor	Claude Opus 4.8 (Anthropic), invoked via Claude Code
Date	2026-07-28

The update is confirmed **editorial-only**, it **accurately resolves** the single open observation F-G05, and it changes nothing in the mathematics or the Lean development. The cumulative Paper III status is upgraded from PASS_WITH_OBSERVATIONS to **PASS**. The verdict is earned: every "did-not-change" claim was checked by byte-level diff and SHA-256, and the new §11.6 text was checked against the axiom footprint measured directly in the main audit.

1. Manuscript diff — only §11.6 wording + version header

A full `diff` (2175 → 2177 lines) shows exactly three edits: (i) the version-header wording; (ii) the §11.6 sparse-node table row, corrected from "proved, relative to AX2" to "the exposed node E_8 depends on AX1 and AX2; the very-sparse core lemma is AX2-only"; and (iii) a new paragraph stating the regime labels summarize the mathematical architecture, and that E_8 dispatches an intermediate alpha-range through the bulk lemma E_4_3 and therefore inherits AX1. No theorem statement, definition, constant, displayed identity, or proof step changed.

2. The correction is factually accurate

The main audit measured, via `#print axioms`, that `PaperIII.E_8` depends on `[propext, Classical.choice, AX1, AX2, Quot.sound]`, the AX1 arising from the E_4_3 dispatch on $\alpha \in [1/12, 1/2]$ while the very-sparse core `E_8_very_sparse_packing_estimate` ($12q < p$) is AX2-only. The v1.1.6 text states exactly this. **F-G05 is resolved.**

3. Lean source and frozen archive are byte-identical

No new freeze was created. Seven load-bearing sources re-hash to their recorded values in `FREEZE_MANIFEST_SHA256.txt` (all MATCH); no .lean file is newer than the manifest. Because the Lean is unchanged, every axiom-gate result and CONFIRMED finding from v1.1.5 carries over verbatim — no rebuild required.

Frozen source	SHA-256 (actual = manifest)
`PaperIII/AX.lean`	MATCH
`PaperIII/E_8.lean`	MATCH
`PaperIII/E_4_3.lean`	MATCH

Frozen source	SHA-256 (actual = manifest)
`PaperIII/Main.lean`	MATCH
`PaperIII/Prop_10_1.lean`	MATCH
`PaperIII/Duality.lean`	MATCH
`PaperIII/Defs.lean`	MATCH

4. Status comparison

Item	v1.1.5	v1.1.6
Theorem 1.1 / Corollary 1.2 sound, conditional on AX1+AX2	PASS	PASS (unchanged)
AX1/AX2 faithful to literature, not overstated	PASS	PASS (unchanged)
Corridor genuinely unconditional (closed footprints)	PASS	PASS (unchanged)
§11.6 sparse-node dependency matches Lean	F-G05 (minor)	**resolved**
Global verdict	PASS_WITH_OBSERVATIONS	**PASS**

Transparency note (not a finding)

The regime-summary sentences at lines 1820/1822 retain the “sparse–AX2” shorthand, now explicitly defined as an architectural summary by the added paragraph; the authoritative node-level table row is correct, so the document is internally consistent and non-misleading. The prior severity-none note F-B02 (AX1 bundles finite LP strong duality with Haxell–Rödl, disclosed by the authors) was always informational and never blocking.

Bottom line: the delta is editorial-only, correct, and complete. Paper III v1.1.6: PASS.

Companion result file SHA-256 (DELTA_AUDIT_RESULT_v1.1.6.md):
`de6e891ad37604b5a1d8b7ad478d8a4f38eecf4534d43431e8611edd5157c881`